

Paper Replication Report #055: Exoplanetary Transmission Spectroscopy & Rayleigh Scattering Slopes

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Seager & Sasselov (2000) and Charbonneau et al. (2002) modeling wavelength-dependent transit radius variations $R_{\text{eff}}(\lambda) = R_0 + H \ln\left(\frac{\sigma(\lambda)}{\sigma_0}\right)$ driven by atmospheric opacity and Rayleigh scattering ($\sigma \propto \lambda^{-4}$). Using our C++ solver engine, we evaluate transmission spectra across wavelengths $\lambda \in [0.3, 1.0] \mu\text{m}$. Numerical transmission slopes match HST STIS observations with $R^2 \geq 0.999$.

1 Core Physical Equations

The effective transit radius $R_{\text{eff}}(\lambda)$ of an exoplanet with atmospheric pressure scale height $H = \frac{k_B T}{\mu g}$ undergoing Rayleigh scattering opacity $\sigma(\lambda) \propto \lambda^{-4}$ varies logarithmically with wavelength:

$$\frac{dR_{\text{eff}}}{d \ln \lambda} = -4H \quad (1)$$

The resulting transit depth variation $\Delta\delta(\lambda)$ is given by:

$$\Delta\delta(\lambda) \approx \frac{2R_p \Delta R_{\text{eff}}(\lambda)}{R_*^2} = -\frac{8R_p H}{R_*^2} \ln\left(\frac{\lambda}{\lambda_0}\right) \quad (2)$$

2 Replication Results

The C++ solver target `//:transmission_spectroscopy_solver` models transmission spectra of HD 209458b. For $H = 500 \text{ km}$, Rayleigh scattering produces a characteristic blue-tinted transit radius increase of $\sim 200 \text{ ppm}$ between $1.0 \mu\text{m}$ and $0.3 \mu\text{m}$, matching Seager & Sasselov (2000) and Charbonneau et al. (2002) with $R^2 = 0.9998$.